ELEC4631 Lecture 12: Controller and Observer Synthesis Using Linear Matrix Inequalities

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LMIs for system analysis and controller synthesis

- Basic examples of how LMIs can be used
  - To check if a certain positive number is a convergence rate for a linear system.
  - Using LMIs for state-feedback design
  - Using LMIs for observer design.

LMI for asymptotic stability

- Consider the simple one state system

$$\dot{x}(t) = -ax(t)$$

- If $a > 0$ then system asymptotically stable and

$$x(t) = e^{-at}x_0 \Rightarrow |x(t)| \le |x_0|e^{-at} \;\; \forall t \ge 0$$

LMI for asymptotic stability

- Consider the simple two state system

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -a & 0 \\ 0 & -b \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

- If $a > 0$, $b > 0$ then system asymptotically stable and

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} e^{-at} & 0 \\ 0 & e^{-bt} \end{bmatrix} \begin{bmatrix} x_{10} \\ x_{20} \end{bmatrix} \Rightarrow \left\| x(t) \right\| \leq \left\| x_0 \right\| e^{-\min(a,b)t} \quad \forall t \geq 0$$

LMI for asymptotic stability

- In general, if linear system

$$\dot{x}(t) = Ax(t)$$

is asymptotically stable then there are constants $C(x_0) > 0$ and $\mu > 0$ such that

$$\|x(t)\| \le C(x_0)e^{-\mu t} \quad \forall t \ge 0$$

LMI for asymptotic stability

- Constant $\mu > 0$ in

$$\|x(t)\| \leq C(x_0)e^{-\mu t} \quad \forall t \geq 0$$

  gives a convergence rate for state to go to 0. That is, how fast state goes to 0.
- Largest value $\mu > 0$ satisfying the above would give fastest convergence rate for system.

LMI for asymptotic stability

- How to check if a number $\mu > 0$ is a convergence rate for system?
- That is, how to check if for a given $\mu > 0$ there is $C(x_0) > 0$ such that

$$\|x(t)\| \leq C(x_0) e^{-\mu t} \quad \forall t \geq 0$$

- We will use LMI approach (useful for later use).

LMI for asymptotic stability

- Observe that if $P = P^T > 0$ then
$$\lambda_{\min}(P)I \leq P \leq \lambda_{\max}(P)I$$
with $\lambda_{\min}(P) > 0$ smallest eigenvalue of P and $\lambda_{\max}(P) > 0$ largest eigenvalue of P.
- Explanation in-class.

LMI for asymptotic stability

- Suppose that there is $P = P^T$ such that

$$P > 0$$

$$A^T P + PA \leq -2\mu P$$

so system is asymptotically stable.

- Consider Lyapunov function $V(x) = x^T P x$. Then

$$\dot{V}(x(t)) = x(t)^T (A^T P + PA) x(t)$$

$$\leq -2\mu x(t)^T P x(t)$$

LMI for asymptotic stability

- So

$$\dot{V}(x(t)) \le -2\mu x(t)^T P x(t) = -2\mu V(x(t))$$

- By result known as Grönwall's inequality

$$\dot{V}(x(t)) \le -2\mu V(x(t)) \Rightarrow V(x(t)) \le e^{-2\mu t} V(x_0)$$

LMI for asymptotic stability

- But

$$\lambda_{\min}(P)\|x(t)\|^2 \le x(t)^T P x(t) \le \lambda_{\max}(P)\|x(t)\|^2$$

- So

$$\lambda_{\min}(P)\|x(t)\|^2 \le V(x(t)) \le \lambda_{\max}(P)\|x(t)\|^2$$

- Therefore

$$\lambda_{\min}(P)\|x(t)\|^2 \le V(x(t)) \le e^{-2\mu t}V(x(0)) \le e^{-2\mu t}\lambda_{\max}(P)\|x_0\|^2$$

LMI for asymptotic stability

- Conclusion is

$$\lambda_{\min}(P)\|x(t)\|^2 \leq \lambda_{\max}(P)\|x_0\|^2 e^{-2\mu t} \Leftrightarrow \|x(t)\| \leq \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}}\,\|x_0\|e^{-\mu t}$$

- That is, μ is a convergence rate for $x(t)$ going to 0 with $C(x_0) = (\lambda_{\max}(P)/\lambda_{\min}(P))^{1/2}\|x_0\|$.

LMI for asymptotic stability

- Summary: if for a given $\mu > 0$, the LMI

$$P > 0$$

$$A^T P + PA \leq -2\mu P$$

is feasible then μ is a convergence rate for the state to go to 0.

LMI for asymptotic stability

- Consider the linear system

$$\dot{x}(t) = Ax(t)$$

- From last week: if the LMI in $P = P^T$

$$P > 0$$

$$A^T P + PA < 0$$

is feasible then system asymptotically stable.

LMI for stabilising feedback

- Consider now controlled system

$$\dot{x}(t) = Ax(t) + Bu(t)$$

- If use state-feedback $u(t) = -Kx(t)$, closed-loop system is

$$\dot{x}(t) = (A - BK)x(t).$$

LMI for stabilising feedback

- If (A,B) controllable then there is K such that

$$\dot{x}(t) = (A - BK)x(t)$$

asymptotically stable (all eigenvalues of A-BK in LHP). Such feedback gain K is stabilising.

- Already know two methods for finding stabilising K: Ackermann's Formula (Lecture 8) and LQR theory (Lecture 9).

LMI for stabilising feedback

- Problem of finding stabilising feedback gain K can also be formulated as LMI problem (this lecture).

LMI for stabilising feedback

- If K is such that closed-loop system

$$\dot{x}(t) = (A - BK)x(t)$$

asymptotically stable then there must exist $P = P^T$ such that

$$P > 0$$

$$(A - BK)^T P + P(A - BK) < 0$$

LMI for stabilising feedback

- The matrix inequalities

$$P > 0$$

$$(A - BK)^T P + P(A - BK) < 0$$

do not form LMI since map

$$L(K,P) = (A - BK)^T P + P(A - BK)$$

not affine.

LMI for stabilising feedback

- Map

$$L(K,P) = (A - BK)^T P + P(A - BK)$$

not affine because it contain products $K^T B^T P$ and PBK involving matrix variables K and P (it is not of form discussed in Week 9).

LMI for stabilising feedback

- But from past lectures we know that if (A,B) controllable there must exist P and K satisfying

$$P > 0$$

$$(A - BK)^T P + P(A - BK) < 0$$

- By introducing new variables we can transform above inequalities into LMI.

LMI for stabilising feedback

- First of all recall: $P > 0 \Leftrightarrow P^{-1} > 0$. Also, $(P^{-1})^T = P^{-1}$.
- Multiplying both inequalities on left and right by P^{-1} yields

$$P^{-1} > 0$$

$$P^{-1}(A^T - K^T B^T) + (A - BK)P^{-1} < 0$$

LMI for stabilising feedback

- Introduce new variables $X = P^{-1}$ and $Y = KP^{-1}$. $X = X^T$ since $P = P^T$ symmetric, but Y not.
- K can be recovered from X and Y as $K = YX^{-1}$.
- Inequalities now become

$$X > 0$$

$$XA^T - Y^T B^T + AX - BY < 0$$

LMI for stabilising feedback

- The map

$$L(X,Y) = XA^T - Y^T B^T + AX - BY$$

affine in X and Y (X symmetric, Y not).

- So inequalities

$$X > 0$$

$$XA^T - Y^T B^T + AX - BY < 0$$

is LMI in $X = X^T$ and Y.

LMI for stabilising feedback

- If $X = X^T$, Y solves LMI

$$X > 0$$

$$XA^T - Y^T B^T + AX - BY < 0$$

then feedback gain K is $K = YX^{-1}$.

Example: LMI for stabilising feedback

- Suppose

$$A = \begin{bmatrix} 0 & 4 \\ -1 & 0 \end{bmatrix}, \; B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(A,B) is controllable and A marginally stable.
LMI

$$X > 0$$

$$XA^T - Y^T B^T + AX - BY < 0$$

has a solution.

Example: LMI for stabilising feedback

- With Matlab obtain a solution

$$X = \begin{bmatrix} 1.6857 & -0.3817 \\ -0.3817 & 0.4135 \end{bmatrix}, \; Y = \begin{bmatrix} -0.8747 & -1.0337 \end{bmatrix}$$

$$K = YX^{-1} = \begin{bmatrix} -1.3716 & -3.7661 \end{bmatrix}$$

- Eigenvalues of *A-BK* are -1.1972 $\pm$ *j*3.4379, closed-loop system asymptotically stable.

LMI for stabilising feedback

- We know that new LMI

$$X > 0$$

$$XA^T - Y^T B^T + AX - BY < 0$$

is feasible if (A,B) controllable.

- However, LMI does not specify how fast closed-loop system will go to zero.

LMI for stabilising feedback

- From Lecture 8: if (A,B) controllable then we know that system can be made to go to 0 as fast as desired (at least in theory).
- LMI can be modified for this purpose.

LMI for stabilising feedback

- For desired minimal convergence rate $\mu > 0$, consider now the inequalities

$$P > 0$$

$$(A - BK)^T P + P(A - BK) \leq -2\mu P$$

LMI for stabilising feedback

- Introducing variables X and Y as before, inequalities can be transformed to LMI

$$X > 0$$

$$XA^T - Y^T B^T + AX - BY \leq -2\mu X$$

- If $X=X^T$, Y solution of LMI, $K = YX^{-1}$ as before.

Example: LMI for stabilising feedback with minimum convergence rate

- Again consider

$$A = \begin{bmatrix} 0 & 4 \\ -1 & 0 \end{bmatrix}, \; B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

  and set minimum convergence rate $\mu = 3$.
  Solve LMI

$$X > 0$$

$$XA^T - Y^T B^T + AX - BY \leq -6X$$

Example: LMI for stabilising feedback

- With Matlab obtain a solution

$$X = \begin{bmatrix} 0.9294 & -0.4243 \\ -0.4243 & 0.2222 \end{bmatrix}, \; Y = \begin{bmatrix} 0.1616 & -1.2223 \end{bmatrix}$$

$$K = YX^{-1} = \begin{bmatrix} -18.1923 & -40.2308 \end{bmatrix}$$

- Eigenvalues of A-BK are -8.9160 and -13.1224, closed-loop system has convergence rate at least $\mu = 3$, as desired.

LMI for stabilising feedback

- So, now you know three different methods of computing K.
- Methods: Ackermann's Formula, LQR theory, and LMIs (without or with specification of closed-loop convergence rate).

LMI for Luenberger observer

- Designing a Luenberger observer can also be formulated as LMI problem.
- Recall: observer can be designed separately from feedback gain K (separation principle).
- Observer gain L chosen so that all eigenvalues of A-LC in LHP.

LMI for Luenberger observer

- If (C,A) observable, Luenberger observer can be designed.
- Same as design of K but make substitutions A^T for A, C^T for B, and L^T for K.

LMI for Luenberger observer

- If (C,A) observable, LMI for observer design without convergence rate specification is

$$X > 0$$

$$XA - Y^T C + A^T X - C^T Y < 0$$

- If $X=X^T, Y$ solution of LMI, observer gain $L = X^{-1} Y^T$.

LMI for Luenberger observer

- If (C,A) observable, LMI for observer design with $\mu > 0$ as minimum convergence rate is

$$X > 0$$

$$XA - Y^T C + A^T X - C^T Y \leq -2\mu X$$

- If $X = X^T, Y$ solution of LMI, observer gain $L = X^{-1} Y^T$.